

**HostelHub**

Security & Data Protection Document

Aapka Data Kaise Safe Hai, Server Kahan Hai, Aur Web App Kaise
Mobile APK me Convert Hua (Hinglish Guide)

DOCUMENT TYPE

Security & Infrastructure Guide (Hinglish)

API APPLICATION SERVERRender Cloud Platform (HTTPS Cloud
Instance)**DATABASE HOST SERVER**

Supabase PostgreSQL (AWS Data Center)

WEB TO APP CONVERSION

Capacitor v8 Android Native Bridge

1. Data Kaise Safe Ho Raha Hai (Encryption & Security)

HostelHub me student, warden aur staff ke data ki privacy aur security ke liye 5-layer encryption aur security guards lagaye gaye hain.

 **Password Hashing (Zero Plaintext Storage):** Aapka login password database me kabhi bhi real text me save nahi hota. Password ko **bcryptjs algorithm (10 salt rounds)** ke dwara highly secure hash string me convert karke save kiya jata hai. Isko decode karna impossible hai.

SECURITY MECHANISM	KAUN SA TECH USE HUA	AAPKA DATA KAISE SECURE HOTA HAI
Password Protection	bcryptjs (10 Salt Rounds)	One-way hashing ke dwara passwords ko unreadable hash string me store karta hai.
API Authentication	JWT Bearer Tokens	Har API request ke sath encrypted token verify hota hai jisse unauthorized log access na kar sakein.
Data in Transit	HTTPS / SSL Encryption	Mobile app aur server ke bich hone wala saara data HTTPS SSL encryption se lock rehta hai.
Database Safety	Sequelize ORM Safe Queries	SQL Injection attacks ko fully block karta hai. Raw SQL injection harmful code run nahi ho sakta.
Security Headers	Helmet.js Security Guard	XSS attacks, MIME sniffing aur unauthorized cross-origin requests ko express server par reject karta hai.

2. Data Kahan Aur Kis Server Par Store Ho Raha Hai?

Cloud Server Details:

- **Database Server:** System ka saara database (Users, Complaints, Mess Menu, Attendance, Group Chats) **Supabase Cloud Managed PostgreSQL Server** par store ho raha hai. Supabase world's most secure **Amazon Web Services (AWS) Data Centers** par run hota hai.
- **Backend REST API Server:** Server logic aur controllers **Render.com Cloud App Hosting Platform** (<https://hostelhub-0cyi.onrender.com>) par hosted hain.
- **Photo & File Uploads:** Complaint issue images, work completion proofs, aur profile pictures Cloud Storage Buckets par base64 / encrypted storage paths ke dwara fully secure rehte hain.

3. Ye Website Kisne Aur Kaise App me Convert Kiya?

HostelHub web application ko **Capacitor v8 (Native Android Bridge Framework)** ki madad se high-performance Android APK me convert kiya gaya hai.

3.1 Conversion Process Kaise Work Karta Hai:

1. **Angular Single-Page App Build:** Pehle Angular framework ke through code (HTML/CSS/TypeScript) ko fast web application bundle me build kiya jata hai.
2. **Capacitor Native Sync:** Capacitor bridge tool (`npx cap sync`) web build folder ko Android native project directory (`android/app/src/main/assets/public`) me insert karta hai.
3. **Android Native WebView Engine:** Android OS ke andar pre-built native Chrome WebView container web application ko 60 FPS full screen mobile app ki tarah execute karta hai. Isme browser ka address bar, URL, ya controls bilkul hide ho jate hain.
4. **Native Mobile Hardware Integrations:**
 - `@capacitor/camera`: Direct mobile camera hardware se issue picture clicked photo upload karne me madad karta hai.
 - `@capacitor/preferences`: Android SharedPreferences me login token save rakhta hai jisse app restart hone par bhi login lost na ho.
 - `@capacitor/app`: Android Hardware Back Button click event capture karta hai aur exit confirmation modal trigger karta hai.

4. Special Profile Security Rule

🔹 **Gender & Batch Critical Edit Warning:** Student jab profile edit karta hai to Name, Room Number, Phone, Bio, ya Profile Picture bina kisi extra approval ke instantly save ho jata hai. Par agar koi student **Gender** ya **Academic Batch** change karega, to system usko critical data warning alert deta hai aur Warden Re-approval (`reApprovalStatus = true`) compulsory kar deta hai.